



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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Abstract

Modern software systems, particularly those built on microservices and containerized architectures, are prone to runtime anomalies such as memory leaks, latency spikes, service crashes, and cascading failures. Traditional monitoring solutions are largely reactive, requiring manual intervention from engineers to diagnose and resolve issues — a process that increases downtime and operational overhead. This project proposes an Adaptive AI-Powered Self-Healing Software Engineering Platform that autonomously detects, diagnoses, and remediates software faults in real time with minimal human intervention.

The platform continuously collects system telemetry — metrics, logs, and traces — using Prometheus, Grafana, and OpenTelemetry, and stores this data in PostgreSQL and Redis for both historical analysis and low-latency access. A machine learning layer, built with Scikit-learn and Pandas, analyzes this telemetry to detect anomalies and predict potential failures before they escalate. Upon detecting an issue, a Large Language Model (via Ollama/Llama 3 or the OpenAI API) reasons over the anomaly context to generate a diagnosis and propose a remediation strategy — such as restarting a pod, scaling a deployment, or rolling back a faulty release.

The system is deployed on Kubernetes (Minikube), orchestrating containerized microservices managed through Docker, with a FastAPI backend exposing the platform's core logic and a React.js and Tailwind CSS frontend providing real-time visibility into system health and healing actions. GitHub Actions automates continuous integration and deployment, ensuring that remediation logic itself remains testable and version-controlled.

By combining observability, predictive analytics, and generative AI-driven decision-making, the platform demonstrates a self-healing feedback loop that reduces mean time to recovery (MTTR), minimizes manual DevOps intervention, and improves overall system resilience — offering a scalable blueprint for autonomous software operations.

Signature of the Guide

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